

Erdős Problem #669

EP-669

L1 - TRACTABLE

PARTIALLY SOLVED

Geometry

ORIGINAL PROBLEM STATEMENT

Let $F_k(n)$ be minimal such that for any n points in $\mathbb{R}^2$ there exist at most $F_k(n)$ many distinct lines passing through at least k of the points, and $f_k(n)$ similarly but with lines passing through exactly k points.

Estimate $f_k(n)$ and $F_k(n)$ - in particular, determine $\lim F_k(n)/n^2$ and $\lim f_k(n)/n^2$.